Introduction to Code-based Cryptography

October 29, 2026

Blockstream Research

research.blockstream.com

github.com/BlockstreamResearch/zk-book



Plan

1 Error-Correcting Codes

2 Reed–Solomon Codes

3 Proximity to a Code

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Goal

Linear codes, Reed–Solomon codes $\mathrm{RS}[\mathbb{F}, \Omega, k]$, and the notion of **proximity** to a code — the central tool behind STARKs.

Error-Correcting Codes

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Fix an **alphabet** Σ (say, $\Sigma = \{0, 1\}$ for binary strings).

Problem

Alice wants to send Bob a message $m \in \Sigma^k$. Alice sits in Germany, while Bob is on the Moon. Whatever message Alice picks, two random letters of m get corrupted. How can Alice transmit the exact message m to Bob?

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- A **decoder** recovers m using the redundant information.

Transmission over a Noisy Channel

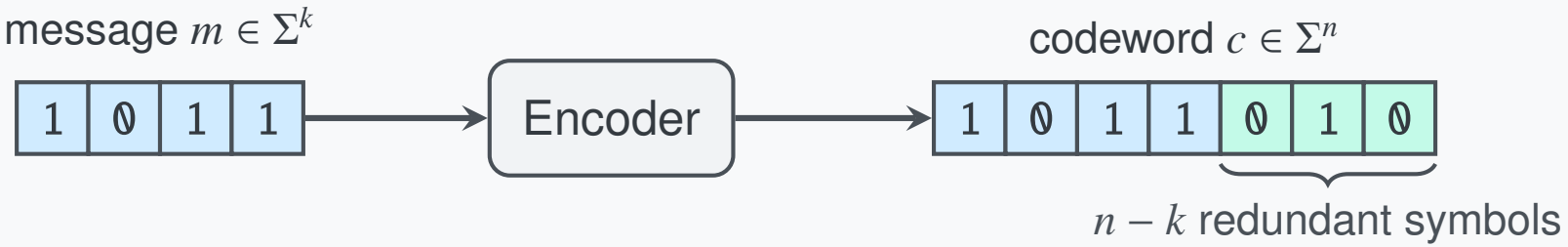


Figure: The encoder expands $k = 4$ symbols into $n = 7$ symbols. The channel corrupts one symbol, and the decoder still recovers the original message.

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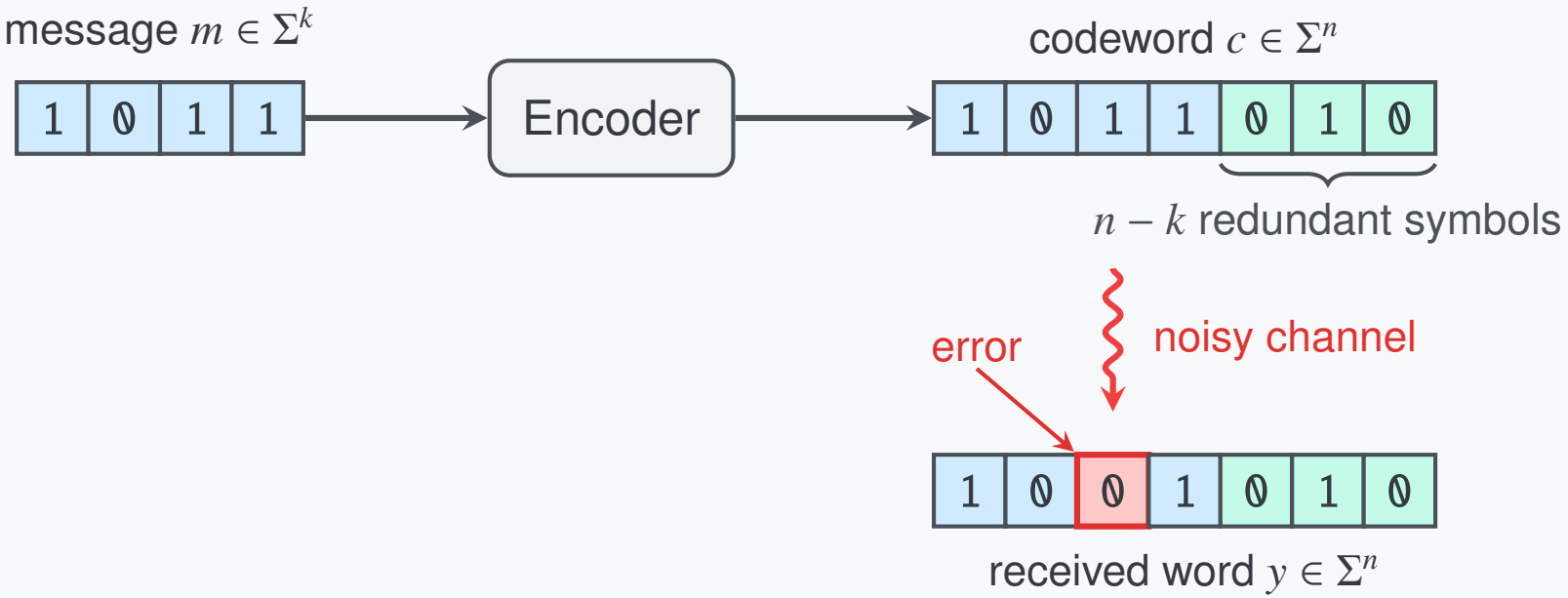


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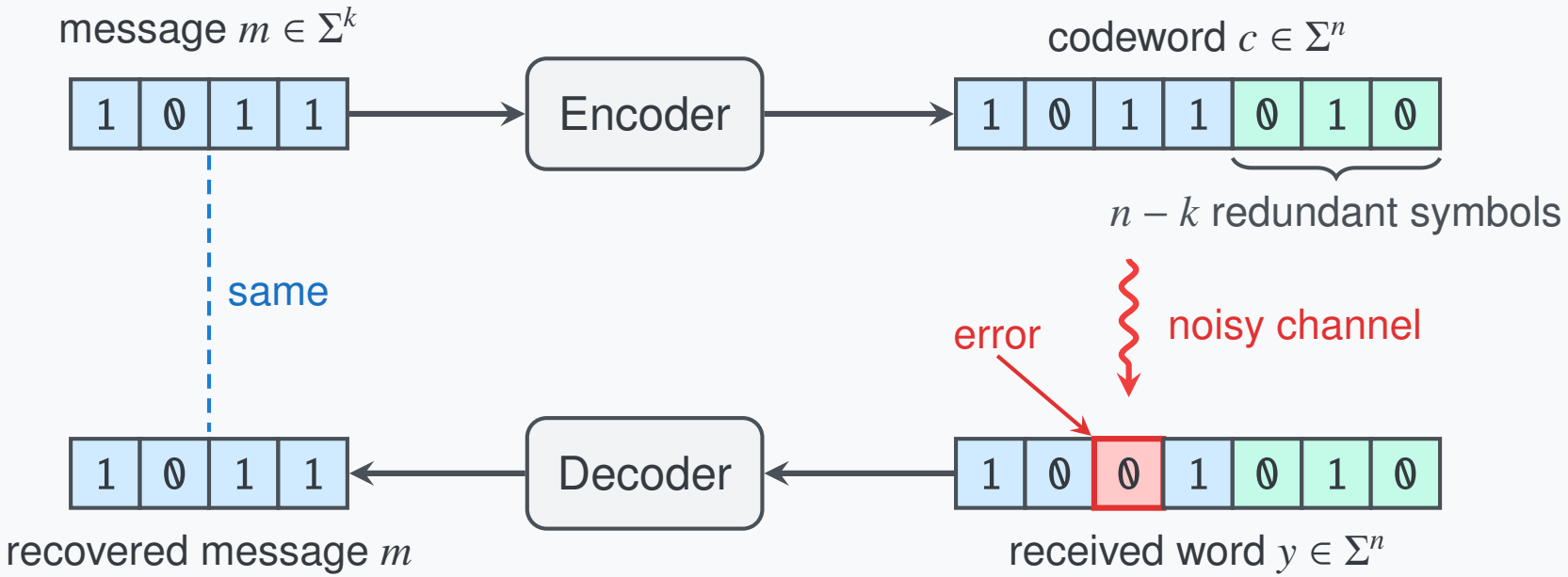


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Code

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A **code** of block length n over an alphabet Σ is a subset $C \subseteq \Sigma^n$, whose elements are called **codewords**.

- $\#C$ is the number of codewords;
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Question

How do we measure "how much" two words differ?

Hamming Distance

Definition (Hamming distance)

Given $\mathbf{x}, \mathbf{y} \in \Sigma^n$, the **Hamming distance** $\Delta(\mathbf{x}, \mathbf{y})$ is the number of positions in which they differ:

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Let $\Sigma := \{0, 1\}$, $\mathbf{x} := 00000$ and $\mathbf{y} := 10100$. Then $\Delta(\mathbf{x}, \mathbf{y}) = 2$: they differ on positions 1 and 3.

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Question #1

Check that (Σ^n, Δ) is a metric space: $\Delta(\mathbf{x}, \mathbf{x}) = 0$, positivity, symmetry, and the triangle inequality $\Delta(\mathbf{x}, \mathbf{z}) \leq \Delta(\mathbf{x}, \mathbf{y}) + \Delta(\mathbf{y}, \mathbf{z})$.

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Question #2

Over $\mathbb{F}_3$, let $\mathbf{x} = (1, 0, 2, 2, 1)$ and $\mathbf{y} = (1, 2, 2, 0, 1)$. What are $\Delta(\mathbf{x}, \mathbf{y})$ and $\delta(\mathbf{x}, \mathbf{y})$?

Minimum Distance

A code's effectiveness depends on the whole set: the worst-case spacing between two codewords is the key metric.

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If $d_{\min}(C) = d$, then:

- fewer than d errors never produce another valid codeword;
- with fewer than $d/2$ errors, the corrupted word is strictly closer to the original codeword than to any other.

Unique Decoding

Proposition

Let $C \subseteq \Sigma^n$ be a code. For every $\mathbf{w} \in \Sigma^n$ there exists at most one codeword $\mathbf{x} \in C$ such that $\delta(\mathbf{x}, \mathbf{w}) < \mu(C)/2$.

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Proof. If $\mathbf{x} \neq \mathbf{x}'$ were both that close, then

$$\delta(\mathbf{x}, \mathbf{x}') \leq \delta(\mathbf{x}, \mathbf{w}) + \delta(\mathbf{w}, \mathbf{x}') < \mu(C),$$

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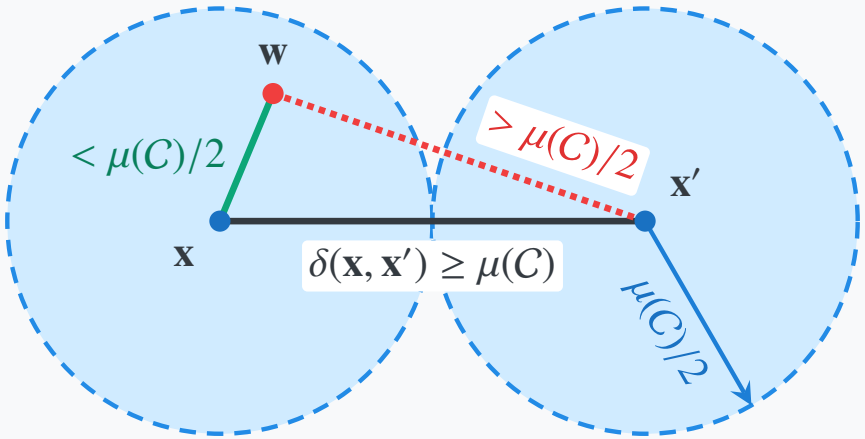
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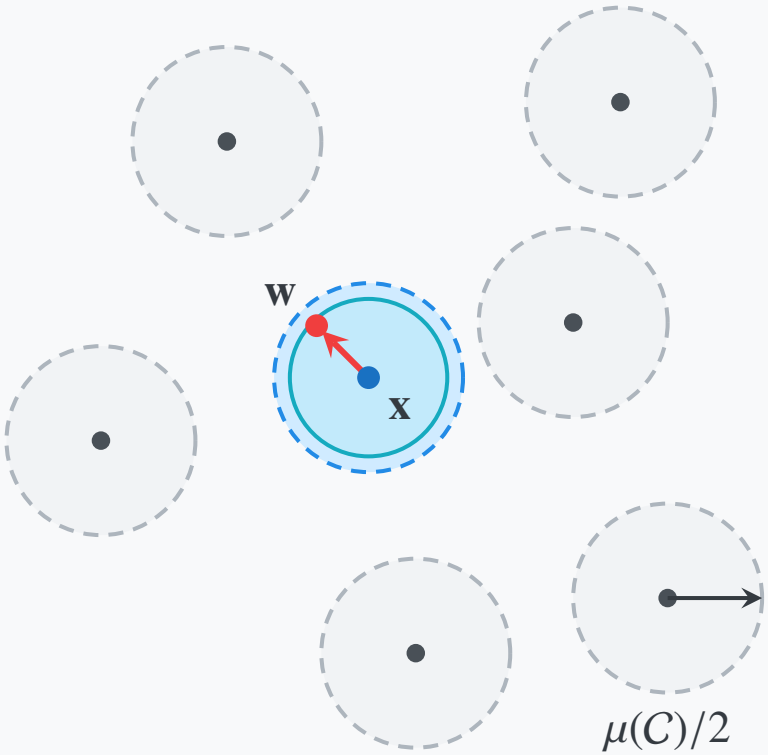
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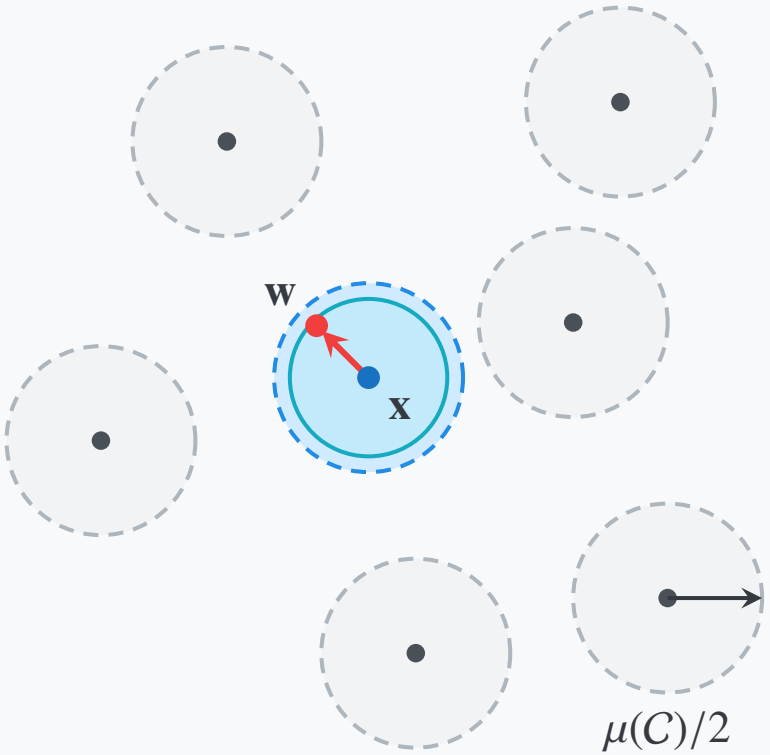
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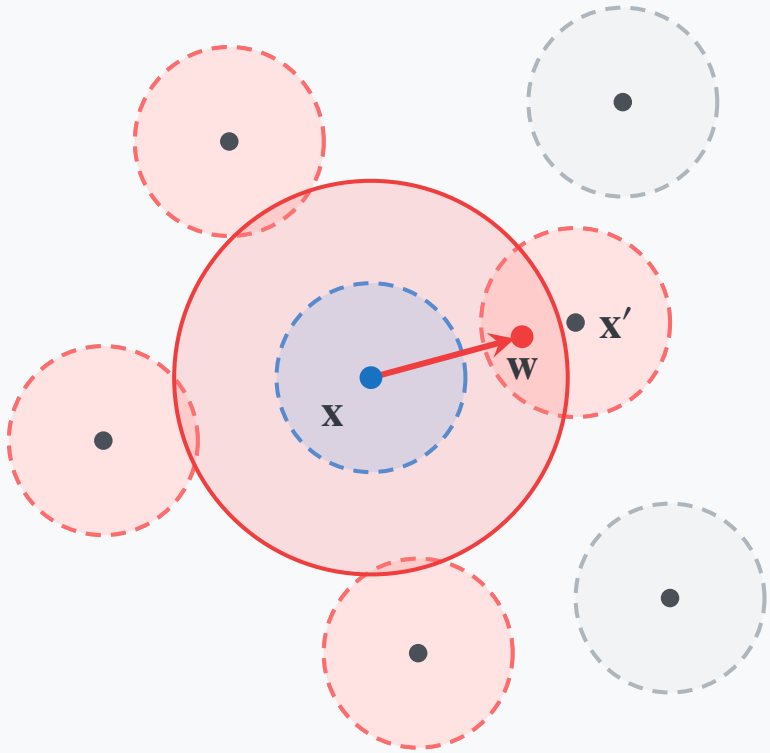
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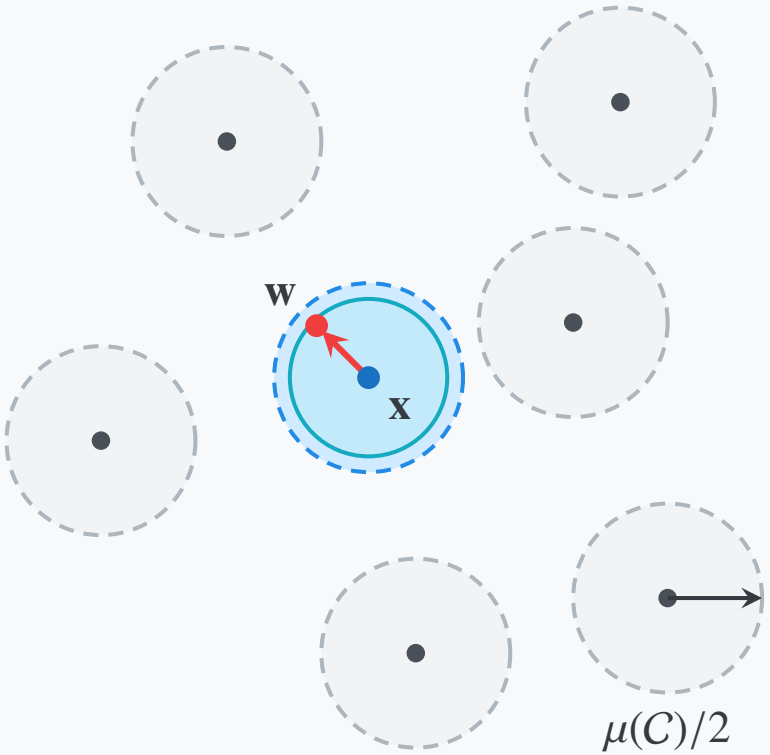
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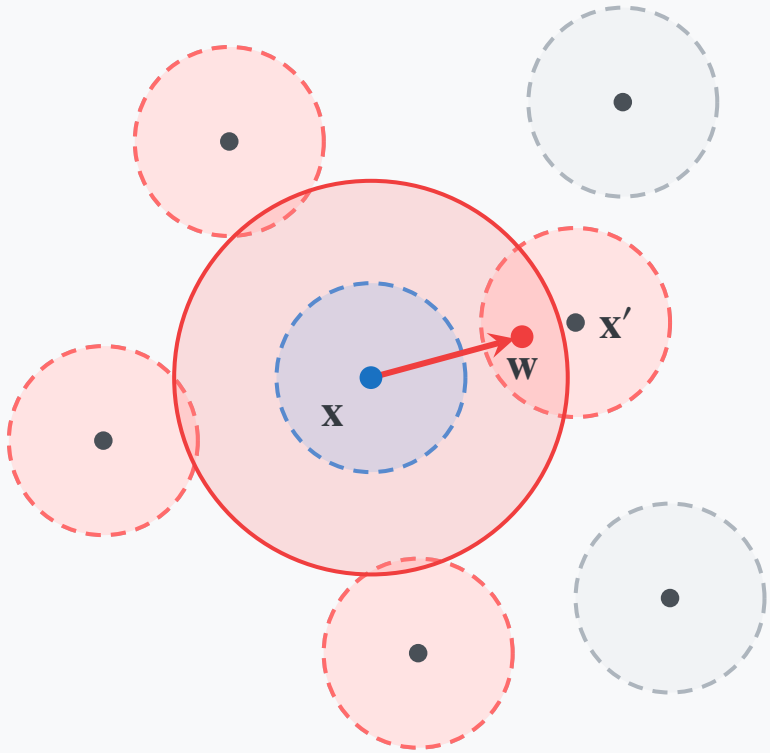
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$\mu(C)/2$ is called the **unique decoding radius** of C.

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- ✓ One bit flipped, $000 \mapsto 100$: still closer to 000 than to 111, since $d/2 = 3/2 > 1$.
- ✗ Two bits flipped, $000 \mapsto 110$: closer to 111, decoding fails, since two errors are not fewer than $d/2$.

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When Σ is a field,

$$\Delta(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_0,$$

that is, two vectors differ exactly in the coordinates where their difference is non-zero.

Linear Codes

Cryptographic codes are never arbitrary subsets of Σ^n: they have algebraic structure.

Definition (Linear code)

Let $\mathbb{F}_q$ be a finite field. A code $C \subseteq \mathbb{F}_q^n$ is a **linear code** if it is a linear subspace of $\mathbb{F}_q^n$. If C has dimension k and minimum distance d, we call it an $[n, k, d]_q$ code.

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Question #3

Show that any linear encoding $\mathbb{F}^k \to C$ is $\mathbf{x} \mapsto \mathbf{G}\mathbf{x}$ for some $\mathbf{G} \in \mathbb{F}^{n \times k}$. What is rank($\mathbf{G}$)?

Distance Equals Weight

The difference of two codewords of a linear code is again a codeword, so the minimum over pairs collapses into a minimum over single codewords.

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- Rate and distance pull in opposite directions: every symbol is spent either on carrying the message or on keeping codewords apart.

Example: Parity Code

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The **parity code** consists of all vectors with an even number of ones:

$$C = \left\{ \mathbf{c} \in \mathbb{F}_2^n : c_1 + c_2 + \cdots + c_n = 0 \right\}.$$

To encode $\mathbf{m} \in \mathbb{F}_2^{n-1}$, append a **parity bit**. For $n = 4$: $101 \mapsto 1010$, $111 \mapsto 1111$.

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- Hence C is an $[n, n-1, 2]_2$ code with rate $\rho(C) = 1 - 1/n$.
- A single error is **detected**, but not **corrected**: 1110 is at distance 1 from both 1010 and 1111.

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To guarantee distance d, a code must dedicate at least $d - 1$ of its n symbols to redundancy.

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- There are at most q^{n-d+1} such strings, so $q^k = \#C \leq q^{n-d+1}$. □

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Question #4

Which of $[7, 4, 4]_q$, $[7, 4, 5]_q$, $[5, 1, 5]_q$, $[6, 3, 4]_q$ cannot be achieved by any linear code?

Reed–Solomon Codes

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2: Fix an **evaluation domain** $\Omega = (\omega_i)_{i\in[n]}$ with $n > k$.

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1: Put the message $(m_i)_{i\in[k]} \in \mathbb{F}^k$ into the coefficients: $f := \sum_{i\in[k]} m_i T^i \in \mathbb{F}[T]^{<k}$.
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Recall: Shamir's secret sharing

Decoding follows the same idea: if few evaluations are corrupted, we can still interpolate the polynomial f of degree $k - 1$.

Reed–Solomon Code

Restriction: for $f \in \mathbb{F}[T]^{<k}$ and a domain $\Omega \subseteq \mathbb{F}$, we write $\overline{f} : \Omega \to \mathbb{F}$ for f viewed only on inputs from Ω.

Remark

Feel free to treat $\overline{f}$ as the vector $(f(\omega_0), \ldots, f(\omega_{n-1}))$ for $\Omega := (\omega_0, \ldots, \omega_{n-1})$.

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Definition (Reed–Solomon code)

The **Reed–Solomon code** over $\mathbb{F}$, evaluation domain $\Omega \subseteq \mathbb{F}$ (with $\#\Omega = n > k$), and degree k, is the linear code

$$\mathrm{RS}[\mathbb{F}, \Omega, k] := \{\overline{f} : \Omega \to \mathbb{F},\ f \in \mathbb{F}[T]^{<k}\}.$$

Example: RS$[\mathbb{F}_7, \mathbb{F}_7, 3]$

Take $\Omega = \mathbb{F}_7$, $n = 7$, $k = 3$.

- $(1, 2, 3) \mapsto f = 1 + 2T + 3T^2$,
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$$\overline{f} = (1, 6, 3, 6, 1, 2, 2),$$
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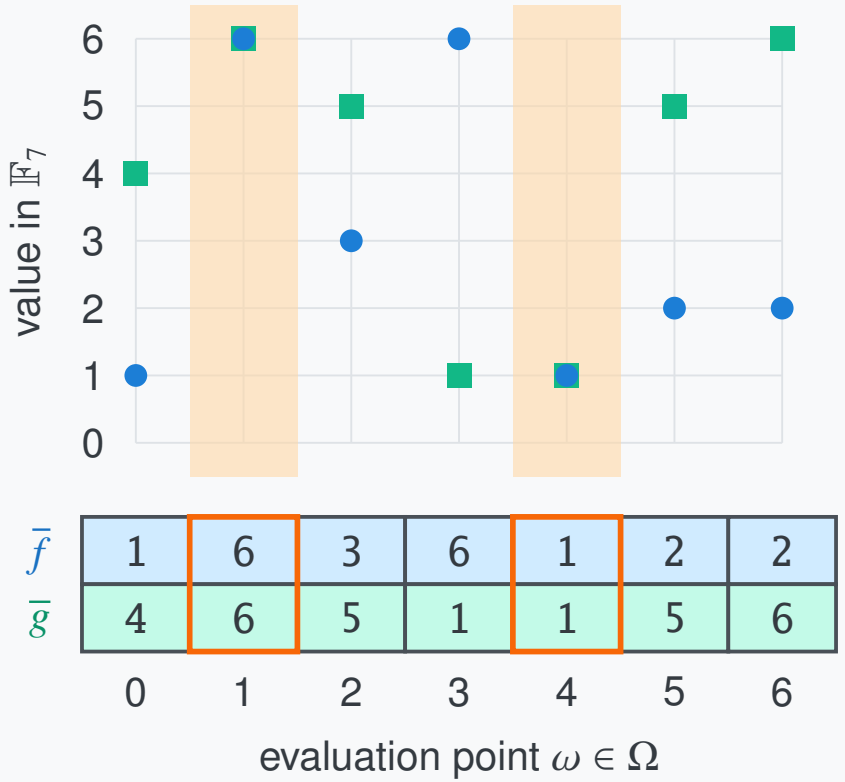
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They agree only at 1 and 4: $f - g = (T - 1)(T - 4)$ has at most $k - 1 = 2$ roots, so $\overline{f}, \overline{g}$ differ in 5 of 7 positions.



Reed–Solomon Code is Linear

Proposition

RS$[\mathbb{F}, \Omega, k]$ is a linear code of length $n = \#\Omega$ and dimension k.

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- The map $f \mapsto \overline{f}$ is linear and injective: if $\overline{f} = 0$, then f has $n > k - 1$ roots, so $f = 0$. Hence the dimension is $\dim \mathbb{F}[T]^{<k} = k$. □

Reed–Solomon Codes are MDS

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$$\Delta(\overline{f}, \overline{g}) = n - \#\{\omega \in \Omega : f(\omega) = g(\omega)\} \geq n - (k-1),$$

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Remark

The rate is $\rho = k/n$ and $\mu = 1 - \rho + 1/n$: lowering the rate directly increases the relative distance. Since n is large in practice, we use $\mu \approx 1 - \rho$.

Questions

Question #5

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Question #6

Let $C \subseteq \mathbb{F}_2^4$ be the linear code spanned by $(1, 1, 0, 1)$ and $(0, 1, 1, 1)$. What is $d_{\min}(C)$?

Proximity to a Code

Distance to a Code

Classical coding theory: recover the original message. Our requirement is weaker:

- the verifier does not need to know **which** polynomial the prover committed to,
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Let $C \subseteq \Sigma^n$ be a code and $\mathbf{w} \in \Sigma^n$. Then

$$\Delta(\mathbf{w}, C) := \min_{\mathbf{x} \in C} \Delta(\mathbf{w}, \mathbf{x}), \qquad \delta(\mathbf{w}, C) := \frac{1}{n} \Delta(\mathbf{w}, C).$$

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$\mathbf{w}$ is δ-**close** to C if $\delta(\mathbf{w}, C) \leq \delta$, and δ-**far** otherwise.

The List of Close Codewords

Definition

Let C be an $[n, k, d]$ linear code. For $\mathbf{w} \in \mathbb{F}^n$ and $\delta \in [0, 1]$, define

$$\text{List}[\mathbf{w}, C, \delta] := \{\mathbf{x} \in C : \delta(\mathbf{x}, \mathbf{w}) \leq \delta\}.$$

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For $\delta \geq \mu(C)/2$ the situation becomes very tricky, and our knowledge is quite limited.

Johnson Bound

Theorem (Johnson bound)

For a code $C \subseteq \mathbb{F}^n$, fixed $\mathbf{w} \in \mathbb{F}^n$, and $0 < \delta < 1 - \sqrt{1 - \mu(C)}$,

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Question #7

Let $C \subseteq \mathbb{F}^n$ be linear over a field of size q, and $\delta \geq \mu(C)$. Show that some $\mathbf{w} \in \mathbb{F}^n$ has $\#\mathrm{List}[\mathbf{w}, C, \delta] \geq q$.

Size of the List

What happens between $1 - \sqrt{1-\mu(C)}$ and $\mu(C)$? There is no general answer: it depends on the specifics of C.

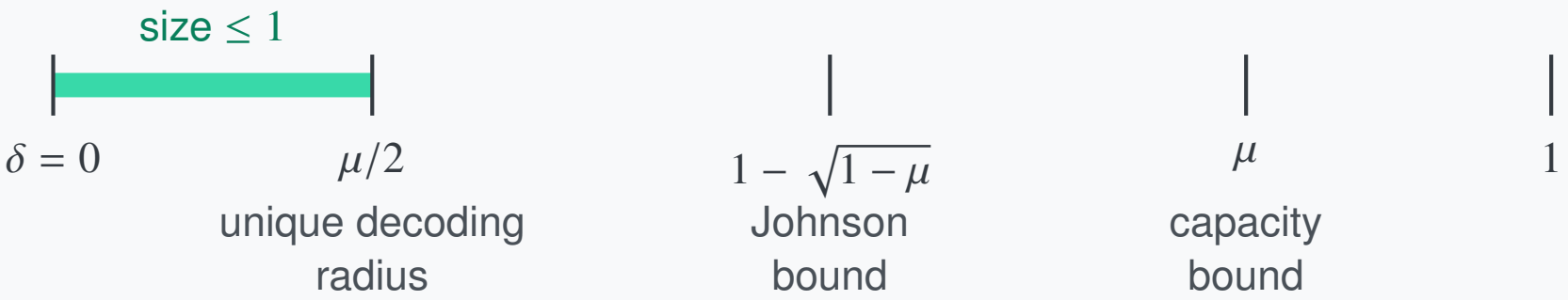


Figure: The size of List$[\mathbf{w}, C, \delta]$ as a function of $\delta \in [0,1]$, where $\mu = \mu(C)$. Adapted from Dan Boneh's ZK Whiteboard Session S2M7.

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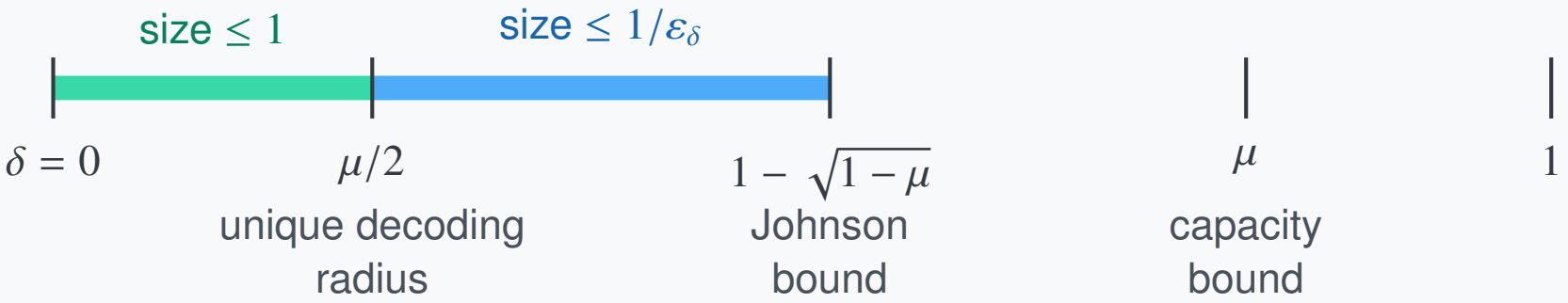


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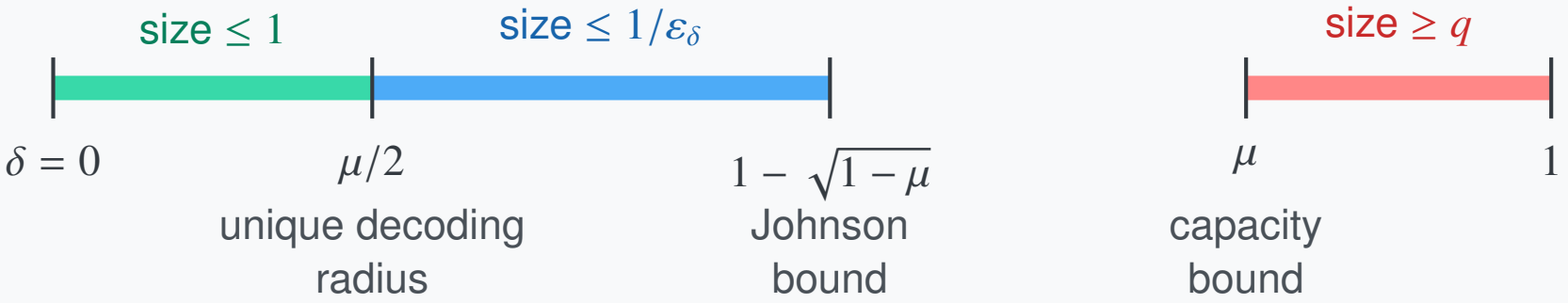


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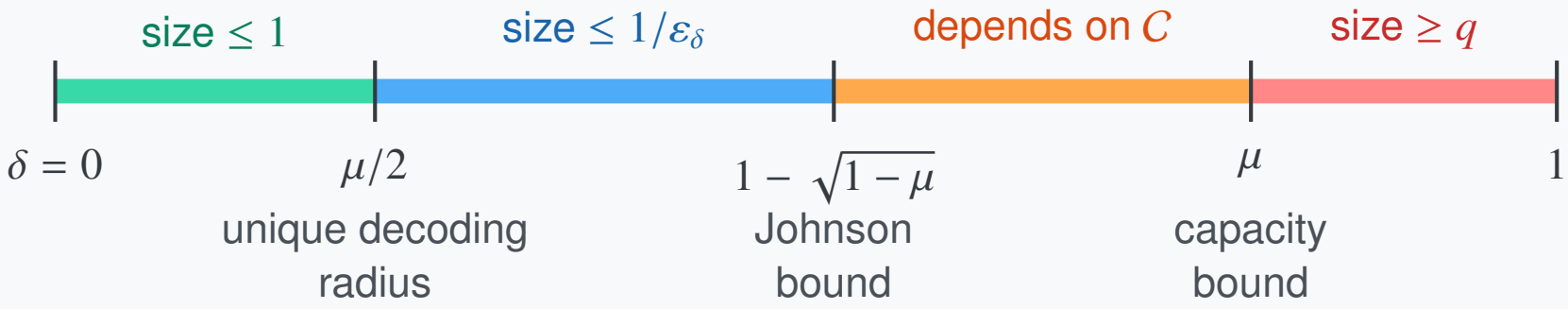


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For $C = \mathsf{RS}[\mathbb{F}, \Omega, k]$, substitute $\mu(C) \approx 1 - \rho(C)$.

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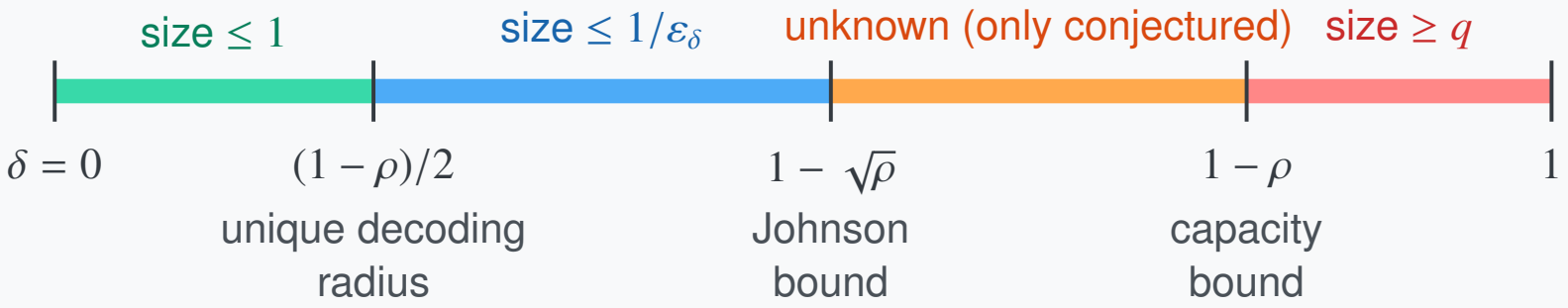
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Question #8

Consider RS$[\mathbb{F}_7, \Omega := \mathbb{F}_7, 2]$ and $\mathbf{w} := (0, 1, 4, 2, 2, 4, 1)$. What is $\delta(\mathbf{w}, \text{RS})$?

Hint: $\mathbf{w}$ is the evaluation of T^2 over $\Omega := \mathbb{F}_7$.

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Assume that $\mathbf{w}$ is δ-far from C.

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The challenge

The verifier does not know in advance which codeword $\mathbf{x}$ it is checking against. An efficient test with few queries to $\mathbf{w}$ is exactly what the FRI protocol achieves.

Three Regimes

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Example

For $\rho = 1/4$ and 100 bits of security, the three regimes require roughly 148, 100, and 50 queries, respectively.

Beyond Proof Systems: Keys from Biometrics

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Let C be a binary $[n, k, 2t+1]_2$ code with a decoder correcting up to t errors, and H a hash function.

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Question #9

Show that $\mathsf{Rep}(\mathbf{h}, \mathbf{w}') = \mathsf{sk}$ whenever $\Delta(\mathbf{w}, \mathbf{w}') \leq t$. For a $[128, 44, 25]_2$ code, why does the key have at most 44 bits of security, however good the biometric is?

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- **Proximity**: the size of List$[\mathbf{w}, C, \delta]$ is ≤ 1 below $(1 - \rho)/2$, small up to the Johnson bound $1 - \sqrt{\rho}$, and only conjectured to be small up to $1 - \rho$.

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- **Linear** $[n, k, d]_q$ codes: distance equals minimum weight; rate $\rho = k/n$; Singleton bound $k \leq n - d + 1$.
- **Reed–Solomon** codes RS$[\mathbb{F}, \Omega, k]$: restrictions of polynomials of degree $< k$ to Ω; they are MDS and $\mu \approx 1 - \rho$.
- **Proximity**: the size of List$[\mathbf{w}, C, \delta]$ is ≤ 1 below $(1 - \rho)/2$, small up to the Johnson bound $1 - \sqrt{\rho}$, and only conjectured to be small up to $1 - \rho$.
- Next: testing proximity with few queries (IOPPs and FRI).

Thank you for your attention



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github.com/BlockstreamResearch/zk-book



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